SIFA KAVEZA MWACHONI

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PROFILE

Software engineer and CTO building data and ML systems for healthcare, water and financial access in East Africa. Seeking to move into technology policy, shaping how digital health and AI reach underserved communities.

EDUCATION

African Leadership University Jan 2023 – Jul 2026

Bachelor of Science in Software Engineering (GPA: 4.26/5.00, over 85%) Kigali, Rwanda

EXPERIENCE

Radah Works Nov 2025 – Present

Chief Technology Officer (CTO) | AI-powered marketplace matching founders with professional teams | Live Demo | GitHub

- As CTO, led technical direction and built the backend and ML matching system with a designer and a frontend developer.
- Automated professional team selection by scoring candidates across 7 criteria, including skills, budget, experience and project fit, generating minimum 4-person, role-diverse teams from project requirements.
- Unified 3 marketplace workflows by building 36 backend API endpoints, Firebase authentication and Stripe payments, bringing founder, professional and administrative operations into one platform.
- Added 118 automated frontend and backend tests and CI checks; every main-branch merge must pass validation.

PROJECTS

WaterPath Kenya | Python, Flask, React, Firebase, scikit-learn, Random Forest, Africa's Talking | Live Demo | GitHub

- Trained a Random Forest on 433 boreholes to predict water quality (68.4% accuracy against a 70.2% majority-class baseline), so I kept predictions out of the app and labelled the 620 untested boreholes "Not tested".
- Expanded access to water point services by adding a USSD prototype (sandbox) alongside the web platform, doubling available access channels and enabling borehole searches and reports without internet.
- Prioritised potential desalination sites by analysing 134 saline/brackish boreholes using location, water quality, population coverage and projected costs, turning raw water data into candidate infrastructure sites.

HealthBridge Facility Finder | Python, Flask, Pandas, NumPy, JavaScript, HTML/CSS, Git | Live Demo | GitHub

- Enabled people to find the nearest health facility for the service they need, across 2,035 location-verified Ugandan facilities, by building and deploying a full-stack Flask app on Render with browser geolocation, haversine distance ranking, and service and facility-type filters.
- Prevented misleading distance results by auditing 6,520 public facility records and finding that 56% had placeholder coordinates, using a pandas pipeline that repairs corrupted text, removes duplicates and out-of-country points, and flags unreliable locations.
- Exposed a coverage bias, showing that excluding unreliable locations would remove 83% of government facilities, by comparing the facility mix before and after and publishing the results on a public data-audit page.

Uwezo KYC | Python, FastAPI, PostgreSQL, React, Docker, PyTorch | GitHub

- Fine-tuned LayoutLMv3 to extract receipt fields (macro F1 0.95 on the SROIE benchmark; 0.82 on synthetic Kenyan receipts, where date extraction was weak at 0.26 F1).
- Built a FastAPI and React receipt-scoring system for alternative KYC; identified privacy risks of using M-Pesa data.
- Limitation: not yet validated on real Kenyan receipts, so real-world performance is untested.

LEADERSHIP AND ACHIEVEMENTS

- Finalist, Raise Her Voice competition (Jan 2025), pitching HealthBridge, a rural and refugee healthcare access platform.
- Developed HealthBridge at the ALX Founder Academy (Nov–Dec 2024).
- Volunteer, ALU matriculation (May 2024) and Code X Treme hackathon (Apr 2024, Apr 2025), Kigali, Rwanda.

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML/CSS
Frameworks: Flask, FastAPI, React
ML and data: PyTorch, scikit-learn, Pandas, NumPy
Tools and platforms: Git, Docker, CI, PostgreSQL, Firebase, Stripe, Render, Africa's Talking (USSD)